



MD QC — Installation and Operation

Automated quality-control monitoring for the Evosep sample-preparation and LC-MS workflow, using the MD QC agent and the Mass Dynamics platform.

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Applies to	MD QC agent v0.5.13 and later	Supersedes	—
Related	Evosep Lupo IFU · Evosep Eno User Manual · MD Platform account documentation		

1. Purpose and scope

This document describes how to install, commission and operate **MD QC** — a monitoring system that evaluates the **Evosep Lupo digest kit and LC-MS performance** by watching an instrument's acquisition folder, extracting quality-control measurements from every QC injection, and tracking them over time against a reference established on that specific instrument.

It covers two components:

- **The MD QC agent** — a background service installed on the instrument PC. Detects new acquisitions, runs a targeted Skyline extraction, and writes a structured result for each run.
- **The Mass Dynamics platform** — the web application where results from one or many instruments are reviewed longitudinally.

In scope

Installation on a Windows instrument PC; configuration for a single instrument; establishing an instrument-specific reference (the *gold standard*); routine interpretation; troubleshooting; and the data-handling questions an IT reviewer will ask.

Not in scope

Operation of the Evosep Lupo, Evosep Eno or the mass spectrometer itself — see the respective Evosep manuals. Preparation of QC A, QC B and SSC₀ samples — see the Evokit instructions. Design of the Skyline method.

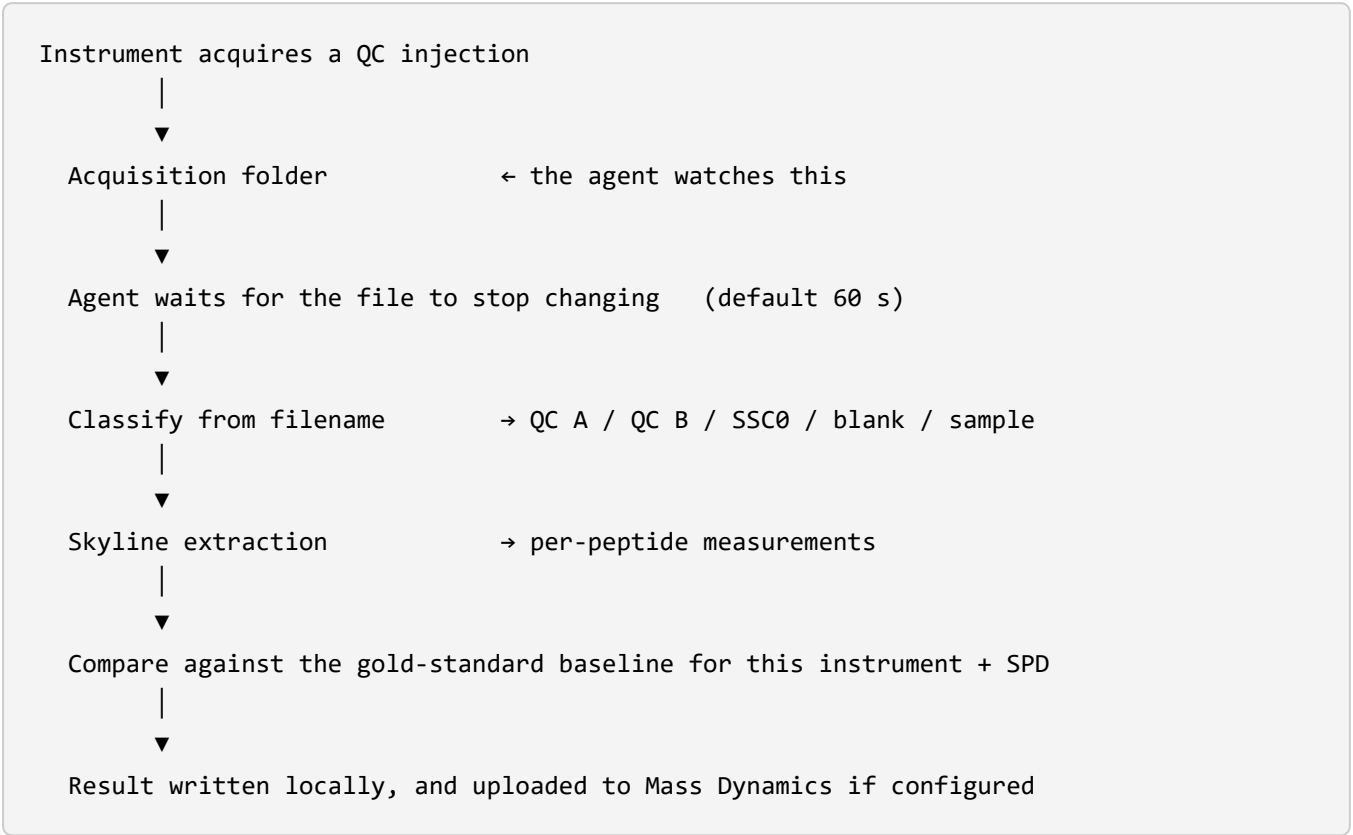
2. Audience and how to use this document

Part	Who	When
Part A	Field service engineer, or a suitably experienced local administrator	Once, at installation, and again after a major instrument service
Part B	Laboratory staff running routine analyses	Daily and weekly, ongoing
Part C	Both, plus IT	As needed

Part A ends with an acceptance checklist (Annex 5) to be completed and left with the customer. Do not hand over to routine operation until every item passes.

3. System overview

The agent runs continuously on the instrument PC and requires no operator action once configured.



The central idea is **comparison against the instrument's own reference**, not against a universal specification. At installation, a set of SSC₀ injections defines how this particular LC-MS and sample-preparation system performs when healthy. Every later QC run is expressed as a ratio to that reference. This is

what makes the same thresholds meaningful across different instruments, and it is why §A9 is a required commissioning step rather than an optional extra.

LOCAL OPERATION

The agent works fully offline. A cloud connection adds longitudinal review across instruments and over time, but is not required for the monitoring loop. Where cloud upload is not permitted, follow Part A but skip §A8, and see §C5 for what this changes.

4. Definitions

Term	Meaning
SSC₀	System suitability control. A standard K562 peptide digest, 50 ng on Evotip, loaded by the Lupo. Used to set the reference baseline for an instrument at installation and after any major change.
QC A	Process control. A lysate carried through the full digestion and Evotip workflow, at roughly 1 µg — about six times the SSC ₀ load on column. Reports on the sample-preparation process.
QC B	Handling control. Materially identical to SSC₀ — the same K562 digest at 50 ng on Evotip, loaded by the Lupo. The difference is how and when each is used: SSC ₀ sets the reference, QC B measures LC-MS performance against it over time. Bypasses digestion, so it reports on the LC-MS and handling rather than on sample preparation.
EB / PB	Evotip blank / plate blank. Injections used to detect carryover.
SPD	Samples per day. The Evosep chromatographic method speed (e.g. 200 SPD). Retention times and intensities differ materially between SPD methods, so each has its own baseline.
Gold standard / baseline	The per-peptide reference values computed from a selected set of SSC ₀ runs on one instrument at one SPD. Also called the <i>local optimum</i> .
Digest efficiency	Ratio of the fully cleaved peptide to the sum of cleaved and miss-cleaved forms, expressed as a percentage.
idotp	Isotope dot product. Agreement between the observed and theoretical isotope pattern. Confidence that the correct precursor was measured.
dotp	Library dot product. Agreement between observed fragment intensities and a spectral library.
Payload	The structured result file the agent writes for one run. See Annex 3.
Spool	The on-disk queue of payloads awaiting upload, and the record of those already sent.
FWHM	Full width at half maximum — peak width. A broadening peak often precedes a drop in signal.

Definitions for individual report metrics will be added once the final metric list is agreed with Evosep.

Installation and commissioning

Performed once, by a field service engineer. Approximately 2–3 hours excluding the SSC₀ run series.

A1. Site and IT prerequisites

Confirm all of the following **before travelling to site**. Several require IT involvement and cannot be resolved on the day.

Requirement	Detail
Operating system	Windows 10 or 11, 64-bit. The instrument's own acquisition PC is the normal host.
Disk space	2 GB free on the system drive. Results are small (~25 KB per run) but Skyline needs working space during extraction.
Skyline	Skyline or Skyline-daily installed, including <code>SkylineCmd.exe</code> . The agent locates it automatically; confirm with <code>mdqc doctor</code> .
Vendor readers	The vendor libraries Skyline needs to read this instrument's raw format must already be present — normally satisfied on an acquisition PC.
Permissions	Read access to the acquisition folder; write access to <code>C:\ProgramData\MassDynamics\QC\</code> . Local administrator rights are required for installation only.
Network (cloud only)	Outbound HTTPS (443) to the Mass Dynamics endpoint. No inbound connection is ever required. See §C5 for the IT summary.
Account (cloud only)	A Mass Dynamics account for the customer's organisation, with an API token available.

CHECK FIRST

Instrument PCs are commonly on a restricted network segment with no outbound internet. Establish this with IT in advance. If egress cannot be granted, the agent still runs — install local-only and revisit later.

A2. Install the agent

Two routes. Use the installer unless the site has a working Python environment and a preference for it.

Route 1 — Windows installer (recommended)

- 1 Obtain the current installer (`mdqc-setup-v0.5.13.exe` or later) from Mass Dynamics.
- 2 Run it as administrator and accept the defaults. The installer registers the agent as a Windows service so it starts automatically with the PC.
- 3 Confirm the service `MassDynamicsQC` appears in Services and is set to start automatically.

Route 2 — Python environment

```
pip install --user --upgrade "git+https://github.com/webwebb56/mdqc-py.git"
```

Then confirm:

```
python -m mdqc version
```

VERIFY BEFORE CONTINUING

Run `python -m mdqc doctor`. It checks the Skyline installation, vendor readers, templates and — if configured — cloud connectivity. Resolve anything it reports before proceeding.

```

MD Local QC Agent - System Health Check
=====

[OK] Agent version: 0.5.9
[OK] Config file: C:\MDQC_Demo\QC\config.toml

Skyline
-----
[OK] SkylineCmd.exe: C:\Program Files\Skyline\SkylineCmd.exe
[--] Skyline version: unknown

Templates
-----
[OK] QC_Method.sky: C:\MDQC_Demo\QC\methods\QC_Method.sky
    Hash: sha256:5c6be2b10a19998b...

Instruments
-----
[OK] Astral_0001: C:\MDQC_Demo\QC\watch: accessible

Cloud
-----
[--] Auth: local-only (no upload)
[OK] Endpoint: https://app.massdynamics.com/api/evosep_qcs
[--] Endpoint reachable: not checked

Spool
-----
[OK] Spool directory: writable
[OK] Pending items: 0
[OK] Failed items: 0

Overall: HEALTHY

```

Figure 1 — Verifying the installation with `mdqc doctor`.

A3. Configure the instrument and watch folder

On first run the agent opens a setup wizard in the browser. If it does not open automatically, the address is printed to the console at startup.

- 1 **Instrument vendor** — select the manufacturer. This determines the file pattern and how the agent decides an acquisition is complete.
- 2 **Instrument ID and watch path** — give the instrument a stable, unique name. This identifies it on the platform and keys its baseline, so choose carefully: changing it later orphans the instrument's history. Set the watch path to the acquisition output folder.

- 3 **Skyline** — confirm the detected `SkylineCmd.exe` path.
- 4 **Skyline template** — see §A4.
- 5 **Output mode** — local-only, or cloud upload. Choose local-only for now; §A8 connects the cloud once extraction is verified.

First-run setup

1. Vendor

2. Instrument

3. Skyline

4. Template

5. Output

Step 1 — Instrument vendor

Select the vendor of the mass spectrometer this agent will watch.

☒ Thermo

☐ Bruker

☐ Sciex

☐ Waters

☐ Agilent

Next

Figure 2 — Setup wizard, step 1 — instrument vendor.

First-run setup

1. Vendor

2. Instrument

3. Skyline

4. Template

5. Output

Step 2 — Instrument ID and watch path

Give this instrument a stable identifier and the directory the agent should monitor.

Instrument ID

Watch path

Likely paths on this machine:

- D:\Data
- D:\Xcalibur\Data
- C:\Xcalibur\Data

Next

Figure 3 — Step 2 — instrument ID and watch path.

First-run setup

1. Vendor

2. Instrument

3. Skyline

4. Template

5. Output

Step 3 — Skyline

Detected SkylineCmd at: C:\Program Files\Skyline\SkylineCmd.exe

SkylineCmd path (leave blank to use detected)

Common locations:

- C:\Program Files\Skyline\SkylineCmd.exe
- C:\Program Files (x86)\Skyline\SkylineCmd.exe

Next

Figure 4 — Step 3 — Skyline location.

First-run setup

1. Vendor

2. Instrument

3. Skyline

4. Template

5. Output

Step 4 — Skyline template

Bundled default: C:\Users\friday\AppData\Local\Temp\claude\C--Users-friday-mdqc-py--claude-worktrees-optimistic-sutherland-
cee532\0f7f038c-02dd-40bf-abc0-47df261a78f1\scratchpad\sop_wizard_data\methods\QC_Method.sky

Custom template path (leave blank to use bundled)

Next

Figure 5 — Step 4 — Skyline template.

First-run setup

1. Vendor

2. Instrument

3. Skyline

4. Template

5. Output

Step 5 — Output mode

☒ Cloud upload (requires API token)

☐ Development — dev.massdynamics.com

☒ Production — app.massdynamics.com

API token

☐ Local-only mode (no cloud auth, payloads stay on disk)

Save and start watching

Figure 6 — Step 5 — output mode and platform environment.

WATCH PATH

Point at the folder the instrument actually writes to, not a synchronised copy or network mirror. The agent needs to observe the file growing and then settling in order to know when acquisition has finished.

A4. Skyline method and report

Two files govern extraction, both in `C:\ProgramData\MassDynamics\QC\methods\`:

File	Role
<code>QC_Method.sky</code>	The Skyline document defining the target peptides and transitions.
<code>MD_QC_Report.skyr</code>	The report definition selecting which columns are exported.

To use a site-specific report, set `report_skyr_path` in the configuration (Annex 1) rather than renaming files. The agent reads the report name from inside the `.skyr`, so the filename does not need to match.

CHECK THE METHOD MATCHES THE INSTRUMENT

Configuring the transitions in `QC_Method.sky` is outside the scope of this document, but before commissioning confirm that the Skyline transition settings match **this instrument platform and its acquisition type** — DDA, DIA, MRM and so on. A method built for a different platform or acquisition mode will extract nothing, and §A7 will fail with zero targets found.

COMMON FAILURE

If the Skyline template is opened in the Skyline GUI and saved with results attached, those embedded replicates travel with the file. The agent strips them automatically at extraction time, but a template distributed with results embedded is much larger and slower to copy. Distribute templates saved without results.

A5. Confirm the file-naming convention

The agent determines what a run *is* from its filename. This is a hard requirement, not a convenience.

REQUIREMENT

Every QC injection filename must contain a control-type marker (`SSC0` , `QCA` , `QCB`) and, where more than one chromatographic method is in use, an SPD marker (`200SPD`). A file with no recognisable control-type marker is classified as a sample and **skipped silently** — no result, no error.

Full specification in Annex 2. To verify against real filenames before relying on them:

```
python -m mdqc classify "SSC0_2026-07-24_200spd_k562_S1-B1.d"
```

Check the reported control type, SPD and confidence. Repeat for one example of each QC type the site will run. Record the results in Annex 5.

```
Classification Result
=====
File: SSC0_2026-08-04_ss_50ng_200spd_k562_qc_s00458_S1-B1_53131.d
Control Type: SSC0
SPD: 200
Dilution: -
Well Position: B1
Instrument: -
Plate ID: -
Confidence: HIGH
Source: FILENAME
```

Figure 7 — Verifying filename classification with `mdqc classify`.

A6. Peptide classes

Peptide classes tell the agent what each target is measuring, so that different peptides are rolled up differently. Configure them to match the Protein column values in the site's Skyline method — see Annex 1.

The most important is the **miss-cleavage pair**, which drives the digest-efficiency figure. It is excluded from target recovery, because the miss-cleaved form is often near the limit of quantitation and counting it would depress recovery artificially.

A7. Single-file verification

A pass/fail gate. Do not continue until this succeeds.

- 1 Copy a single previously acquired QC raw file into the watch folder.
- 2 Wait approximately 90 seconds — a 60 second stability window plus extraction time. Large files take longer.

- 3 Open the agent's Dashboard. The run should appear in recent activity with a target count such as `8/8`.
- 4 Confirm a result file has appeared in `C:\ProgramData\MassDynamics\QC\spool\completed\`.

IF TARGETS READ 0/8

The Skyline method is not matching the acquisition. Open the template in the Skyline GUI, import the same raw file manually and confirm peaks are found there. If they are not, the method or its libraries do not correspond to the acquisition method — for example, libraries built for a different SPD.

A8. Connect to Mass Dynamics

Skip this section for a local-only installation.

DO THIS IN ORDER

Set the retention limit **before** entering the API token. Connecting the cloud changes how long results are kept locally, and a low limit carried over from an earlier installation will begin deleting results as soon as the token is saved.

- 1 Open **Settings** → **Payload retention**. Set **Keep newest** to at least 200, or higher than the largest number of runs expected in a single night. Save.
- 2 Obtain the API token from the customer's Mass Dynamics account (Account Details).
- 3 Open **Settings** → **Cloud**. Select the environment — **Production** for customer installations. Paste the token. Save.
- 4 Restart the agent. A restart is required: the upload component reads its configuration once at startup.
- 5 Confirm upload by dropping a test file and checking that the result moves from `spool\pending\` to `spool\completed\`, and appears on the platform.

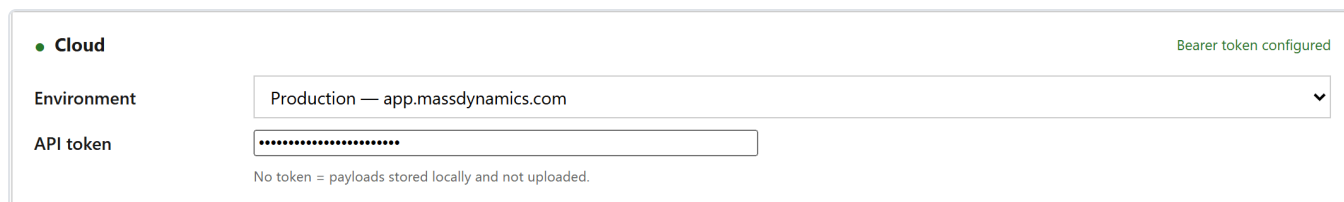
Payload retention Cloud — kept by count

Payloads are uploaded to Mass Dynamics, so `spool\completed\` is a local copy for checking against and is capped at the newest N files.

Keep newest
Maximum payloads retained in `completed\` once cloud upload is configured. A single overnight batch can exceed 150 runs — set this above your busiest expected night.

Keep for (days)
Payloads older than this are removed. In local-only mode this is the only thing bounding `completed\`.

Figure 8 — Settings — payload retention.



● Cloud Bearer token configured

Environment Production — app.massdynamics.com

API token

No token = payloads stored locally and not uploaded.

Figure 9 — Settings — cloud connection. The token field is masked.

IF THE NETWORK IS UNAVAILABLE

Results queue in `spool\pending\` and upload automatically when connectivity returns. Nothing is lost and no operator action is needed.

A9. Record the gold-standard baseline

This step establishes the reference every later run is judged against. It is the core of commissioning.

- 1 **Acquire the SSC₀ series.** Run 15–20 SSC₀ injections at the SPD method the site will use for routine QC. *Evosep's recommended QC method is 200 SPD with the Ev1109 analytical column.* Where more than one SPD will be monitored, repeat for each — baselines are per instrument *and* per SPD.
- 2 **Let the agent process them.** Each is indexed automatically as it completes.
- 3 **Open the Gold standards page** in the agent interface and select the SPD tab.
- 4 **Review the runs.** Each row is one SSC₀ run; columns are the target peptides. Cells are shaded where a value deviates from the current selection, so an outlier stands out.
- 5 **Select the representative runs.** Tick those that should define the reference. The first injection of a series is frequently excluded — a column that has not fully equilibrated shows shifted retention times and depressed areas.
- 6 **Check the resulting baseline.** The lower table shows the per-peptide median and **CV%** across the selected runs. A low CV means the chosen runs agree closely. Investigate before saving if CV% is above about 10% for the recovery peptides.
- 7 **Save.** Give the baseline a label recording why it was made — for example `Insta11 2026-08-04`. Saving activates it immediately for subsequent runs at that SPD.

SSC0 runs

15 of 16 runs selected for baseline

Peak area per peptide, one row per run. Cells are shaded where a value sits outside the spread of the currently selected runs — amber beyond 1 SD, red beyond 2 SD. Colours below identify each peptide and are reused in the retention-time plot.

ISGLIYEETR RISGLIYEETR GDLDAASYAPVR IGGIGTVPVGR LGGEEVSACK PVSSAASYAGAGGSGSR TTPSVVCFK YNSQNSNNQFVLYR

Run	Acquired	Found	ISGLIYEETR	RISGLIYEETR	GDLDAASYAPVR	IGGIGTVPVGR	LGGEEVSACK	PVSSAASYAGAGGSGSR	TTPSVVCFK	YNSQNSNNQFVLYR
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-A1_53100.d	2026-08-12 09:00	6/6	6,933,033	350,875	4,813,950	4,033,392	1,924,325	7,562,850	5,904,500	3,040,133
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-B1_53101.d	2026-08-12 09:07	6/6	6,910,000	349,708	4,797,983	4,020,025	1,917,958	7,658,100	5,884,900	3,030,067
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-C1_53102.d	2026-08-12 09:14	6/6	6,886,967	348,542	4,782,017	4,006,658	1,911,592	7,632,700	5,865,300	3,020,000
☐ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-D1_53103.d	2026-08-12 09:21	6/6	4,255,639	215,372	2,954,951	2,475,841	1,181,240	4,716,526	3,624,334	1,866,159
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-E1_53104.d	2026-08-12 09:28	6/6	6,950,308	351,750	4,825,925	3,979,925	1,898,858	7,581,900	5,919,200	2,999,867
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-F1_53105.d	2026-08-12 09:35	6/6	6,927,275	350,583	4,809,958	4,030,050	1,922,733	7,677,150	5,899,600	3,037,617
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-G1_53106.d	2026-08-12 09:42	6/6	6,904,242	349,417	4,793,992	4,016,683	1,916,367	7,651,750	5,880,000	3,027,550
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-H1_53107.d	2026-08-12 09:49	6/6	6,881,208	348,250	4,778,025	4,003,317	1,910,000	7,626,350	5,860,400	3,017,483
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-A2_53108.d	2026-08-12 09:56	6/6	6,858,175	352,625	4,762,058	3,989,950	1,903,633	7,600,950	5,840,800	3,007,417
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-B2_53109.d	2026-08-12 10:03	6/6	6,944,550	351,458	4,821,933	4,040,075	1,897,267	7,575,550	5,914,300	2,997,350
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-C2_53110.d	2026-08-12 10:10	6/6	6,921,517	350,292	4,805,967	4,026,708	1,921,142	7,670,800	5,894,700	3,035,100
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-D2_53111.d	2026-08-12 10:17	6/6	6,898,483	349,125	4,790,000	4,013,342	1,914,775	7,645,400	5,875,100	3,025,033
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-E2_53112.d	2026-08-12 10:24	6/6	6,875,450	347,958	4,774,033	3,999,975	1,908,408	7,620,000	5,855,500	3,014,967
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-F2_53113.d	2026-08-12 10:31	6/6	6,961,825	352,333	4,758,067	3,986,608	1,902,042	7,594,600	5,835,900	3,004,900
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-G2_53114.d	2026-08-12 10:38	6/6	6,938,792	351,167	4,817,942	4,036,733	1,895,675	7,569,200	5,909,400	3,042,650
☒ SSC0_2026-08-12_ss_50ng_200spd_k562_S1-H2_53115.d	2026-08-12 10:45	6/6	6,915,758	350,000	4,801,975	4,023,367	1,919,550	7,664,450	5,889,800	3,032,583

Figure 10 — Gold standards — SSC₀ runs. Cells show **peak area** per peptide, one row per run; colours identify the peptide and are reused in Figure 11. Shading marks a value outside the spread of the selected runs — amber beyond 1 SD, red beyond 2 SD. The fourth run has not equilibrated, so it is flagged and left unticked and does not contribute to the baseline.



Figure 11 — Retention time across the series, in minutes across the 200 SPD gradient. The excluded run is dimmed and visibly shifted.

Resulting baseline

Peptide	Median area	Area CV%	Median RT (min)	RT sd (s)	n
ISGLIYEETR	6,915,758	0.4%	5.44	0.4	15
RISGLIYEETR	350,292	0.4%	5.97	0.3	15
GDLDAAASYAPVR	4,797,983	0.4%	4.22	0.4	15
IGGIGTVPVGR	4,016,683	0.5%	3.58	0.4	15
LGGEVSVACK	1,911,592	0.5%	2.41	0.4	15
PVSSAASYAGAGGSGSR	7,626,350	0.5%	1.82	0.4	15
TTPSVVCFK	5,884,900	0.4%	4.86	0.4	15
YNSQNSQNSNQFVLYR	3,025,033	0.5%	3.05	0.4	15

Figure 12 — The resulting baseline — per-peptide medians across the selected runs. Peak-area spread is reported as CV%, retention-time spread as standard deviation in seconds.

WHY THE SELECTION IS MANUAL

An engineer commissioning an instrument knows which runs were affected by equilibration, a bubble, or an intentional adjustment. Automatic selection cannot know this. The CV% column exists to make the judgement evidence-based rather than a guess.

A10. Commissioning acceptance

Complete Annex 5 with the customer present. Every item must pass before handover. Walk the customer through Part B, in particular §B3 and §B7.

Routine operation

Laboratory staff. No daily action is required beyond reviewing results.

B1. What happens automatically

Once commissioned the agent requires no intervention. Every QC injection is detected, extracted, compared against the baseline, and — where configured — uploaded, within a few minutes of acquisition completing. The agent starts with the PC.

Your role is to review the results and act when they indicate a problem.

B2. The agent interface

Open the agent's web interface from the system tray icon. It is served locally and is not reachable from other machines.

Page	Use
Dashboard	Service status, queue counts, recent activity. The everyday view.
Gold standards	Review SSC ₀ runs and the active baseline; re-baseline (§B7).
Settings	Configuration. Changing cloud settings requires a restart.
Diagnostics	The same checks as <code>mdqc doctor</code> , in the browser.
Failed	Runs that could not be extracted, with the reason. Retry from here.
Logs	Recent agent log, for support.

Healthy looks like: status Running; pending count at or near zero; failed count zero; recent activity showing each expected QC injection with full target recovery.

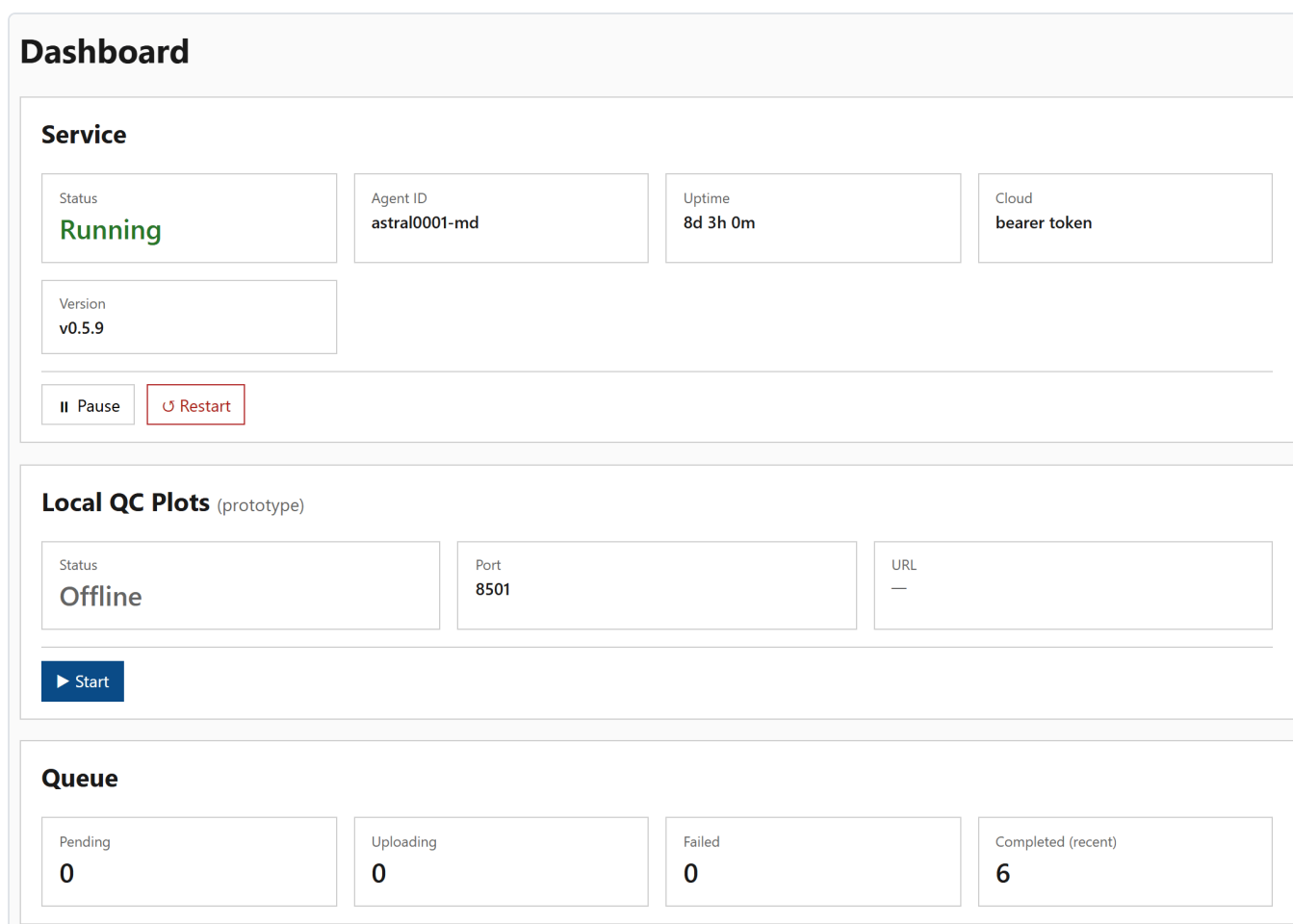


Figure 13 — The agent dashboard in a healthy state.

B2a. Viewing results without the platform

The agent interface shows service health and recent activity, but not the QC measurements themselves. Two options exist for looking at results on the instrument PC:

- **The Gold standards page** — shows every recorded SSC_0 run with its per-peptide values, and the resulting baseline. Useful for a quick check even when you are not re-baselining.
- **The local plots dashboard** — an optional component providing longitudinal trend views on the instrument PC. It is installed with `pip install --user "mdqc[plots] @ git+https://github.com/webwebb56/mdqc-py.git@main"` and started from the Dashboard, or with `streamlit run <install path>/mdqc/plots/app.py`. It reads the stored result files directly and needs no network connection.

WHICH TO USE

The local dashboard is a prototype and is not required for routine operation. Where cloud upload is configured, the Mass Dynamics platform (§B3) is the intended place to review results, and is the only way to compare across instruments or over long periods. Where upload is not permitted, the local dashboard is the only trend view available.

B3. The Mass Dynamics platform

The platform is where results are reviewed over time and across instruments.

- 1 Sign in to your organisation's Mass Dynamics account.
- 2 Open the workspace containing the QC modules.
- 3 **Add your instrument to each module's filter.** Open the module's details panel, select the instrument, and save changes.

MOST COMMON CAUSE OF "MY DATA ISN'T SHOWING"

A module displays only the instruments selected in its filter. A newly commissioned instrument is not selected by default, so its results are uploaded and stored but not displayed until it is added. If data is missing from a view, check this before anything else.

FIG-14, FIG-15

Module details panel with the instrument selector; the same module populated after saving.
Requires a signed-in Mass Dynamics account — capture during the first customer installation.

Figures 14–15 — Adding an instrument to a module.

B4. Reading the QC views

View	Shows	Use it to
Traffic light / scorecard	Recent runs against metrics, coloured by how far each sits from the reference	Answer "is anything wrong right now?" at a glance
Longitudinal tracking	One metric over time, per peptide or aggregated	Distinguish a gradual drift from a single bad run
Ratio panels	QC A and QC B relative to SSC_0	Separate an LC-MS problem from a sample-preparation problem

The ratio panels carry most of the diagnostic value. QC B and SSC_0 are the same material at the same load and both bypass digestion, so QC B relative to SSC_0 reports on LC-MS health. QC A additionally carries digestion, so QC A falling while QC B holds steady points at sample preparation rather than the instrument.

FIG-16, FIG-17

Traffic-light scorecard with a mix of states; longitudinal plot showing a clear drift.
Requires a signed-in Mass Dynamics account with several weeks of data.

Figures 16–17 — Platform QC views.

B5. Interpreting a result

All comparisons are relative to the SSC₀ baseline for that instrument and SPD.

PROVISIONAL VALUES

The thresholds below are a first draft derived from a single instrument. They are configurable and expected to be refined as data accumulates across platforms. Treat them as guidance for investigation, not as an acceptance specification. Current defaults and their origin are in Annex 4.

Is the measurement trustworthy?

Before treating a poor result as an instrument problem, establish that the correct peak was measured. A wrongly integrated peak and a genuine performance drop look identical if only peak area is considered.

Signature	Interpretation
Retention time within 2% of baseline, dot products within 5%	Correct target. Any area change is real. <i>The retention-time limit is being restated as a standard deviation in seconds — see Annex 4.</i>
Retention time beyond 2% <i>together with</i> depressed dot products or a large area change	Probably the wrong peak. Flagged as an extraction problem, not a performance problem.
Retention time shifted, everything else normal	A chromatographic shift affecting all peptides — check for a column or gradient change rather than a single-peptide issue.

DOT PRODUCTS ALONE ARE NOT SUFFICIENT

A wrongly extracted peak has been observed with an isotope dot product of 0.92 — a value that looks healthy. Retention time is the more reliable discriminator, which is why the flag requires a combination of signals.

Is the signal where it should be?

Peak area vs SSC ₀	Indicates
Within about 10%	Normal.
About 10% below	Possibly as much as a 25% reduction in material reaching the column. The response is buffered — a modest area change can reflect a much larger loading change.
About 25% below	Possibly as much as a 50% reduction.
Replicates from one preparation run varying by more than 10%	Inconsistent Evotip loading rather than an instrument problem.

These figures come from a timsTOF HT series, which appears to have the most buffered response of the platforms measured so far. Evosep is producing equivalent data across the other instrument platforms and per recovery peptide — some are more accurate predictors than others — so expect these bands to be refined.

B6. Escalation

Observation	Likely domain	Action
QC B and SSC ₀ both down; QC A down proportionally	LC-MS	Likely a mass-spectrometry issue — perform the MS-specific sensitivity tests recommended by the instrument vendor. For the LC and Evotip side, follow the Evosep troubleshooting guide (https://www.evosep.com/support/troubleshooting/).
QC B and SSC ₀ steady, but QC A recovery peptides down and/or digest efficiency lower	Sample preparation	Review the digestion and Evotip loading for that batch. Consult the Evosep Lupo troubleshooting guide.
QC A digest efficiency down	Sample preparation	Check the kit expiry date, then review digestion conditions for that batch.
Retention times drifting across all peptides, as well as FWHM increase	Chromatography	Check column age and system pressure; follow the Evosep troubleshooting guide (https://www.evosep.com/support/troubleshooting/). Re-baseline after any column change (§B7).
Replicate variability high within one preparation run	Sample preparation	Potential issue with Evotip loading — review the Evosep Lupo troubleshooting guide.
One peptide flagged as an extraction problem, others normal	Data processing	Not an instrument fault. Report to Mass Dynamics — the method may need adjustment.
Runs not appearing at all	Configuration	§C1. Check filenames first (§A5).

B7. Re-baselining

The baseline describes the instrument as it was when recorded. After a deliberate change, it no longer does, and comparisons become misleading.

Re-baseline after: a column change; a major service or clean; a change of chromatographic method; any intentional configuration change expected to alter performance.

Do not re-baseline to make a warning disappear. If performance has degraded unintentionally, a new baseline conceals the problem — the instrument is then measured against its own degraded state.

- 1 Complete the change and allow the system to stabilise.
- 2 Run a fresh SSC₀ series — 15–20 injections at the relevant SPD.
- 3 Open **Gold standards**, select the new runs, and save with a label recording the reason, for example `After column change 2026-09-12`.

Previous baselines are retained. Runs record which baseline they were measured against, so history remains interpretable across a re-baseline.

B8. Routine check schedule

Frequency	Check
Daily	Agent dashboard: running, queue clear, no failures. Traffic-light view for the day's QC injections.
Weekly	Longitudinal view for drift not visible day to day. Digest efficiency trend.
Monthly	Confirm the active baseline still reflects the instrument. Review the failed-files list and clear it.
After any service	Re-baseline per §B7.

PART C

Maintenance, troubleshooting and reference

Laboratory staff, service engineers and IT.

C1. Failure modes

Failure mode	Possible cause	Resolution
No results produced; log repeats <code>SkylineCmd.exe not found</code>	Skyline not installed in a standard location, and no explicit path configured	Install Skyline to its default location, or set <code>[skyline] path</code> in the configuration. Verify with <code>mdqc doctor</code> .
Some QC injections never appear	Filenames lack a recognisable control-type marker, so the runs are classified as samples and skipped	Verify with <code>mdqc classify</code> (§A5). Correct the naming convention, or add a matching rule to the configuration.
<code>0/8 targets found</code> on every run	Skyline method does not match the acquisition method	Import a raw file into the template manually in Skyline. If no peaks are found there, the method or its libraries are wrong for this acquisition — commonly libraries for a different SPD.
Report error naming a report that does not exist	The configured <code>.skyr</code> is missing	Restore the report to the methods folder, or correct <code>report_skyr_path</code> .
Warning that a column override was not found	A column mapping in the configuration refers to a column that a newer report no longer contains	Remove the stale override. The built-in mapping resolves current column names automatically.

Failure mode	Possible cause	Resolution
Skyline error: file in use by another process	Concurrent extractions contending for the chromatogram cache	Handled automatically by per-extraction temporary directories. If it persists, end any stuck <code>SkylineCmd.exe</code> processes and restart the agent.
Only the most recent results remain on disk	Retention limit set too low, or carried over from an earlier installation	Settings → Payload retention . Raise Keep newest above the busiest expected night. See §A8.
Results stay in <code>spool\pending\</code>	No network route to the platform, or an invalid token	Confirm outbound HTTPS. Re-enter the token and restart. Queued results upload automatically once connectivity returns — nothing is lost.
Results upload but do not appear on the platform	The instrument is not selected in the module's filter	Add it (§B3).
Cloud settings saved but nothing uploads	The agent was not restarted	Restart. Cloud configuration is read once at startup; the interface displays a reminder when a restart is required.
No comparison against the baseline in results	No baseline saved for that instrument and SPD, or the SPD could not be read from the filename	Confirm the SPD marker is present (Annex 2), then record a baseline (§A9).

Failure mode	Possible cause	Resolution
A peptide is persistently flagged as an extraction problem	The wrong peak is being integrated, commonly for a low-intensity target	Not an instrument fault. Send a support bundle (§C3) — the method may need adjustment.

C2. File locations

All agent data is under `C:\ProgramData\MassDynamics\QC\`.

Path	Contents
<code>config.toml</code>	Configuration. See Annex 1.
<code>methods\</code>	Skyline template, libraries and report definition.
<code>spool\pending\</code>	Results awaiting upload.
<code>spool\completed\</code>	Results uploaded, or retained locally where cloud upload is not configured.
<code>spool\failed\</code>	Results that could not be uploaded, with the reason.
<code>gold_standards\</code>	SSC ₀ run index and saved baselines. Not subject to routine pruning.
<code>logs\mdqc.log</code>	Agent log.

C3. Collecting a support bundle

Send to support@massdynamics.com:

- ☐ `logs\mdqc.log`
- ☐ `config.toml` — remove the API token before sending
- ☐ Two or three example result files from `spool\completed\`
- ☐ Anything in `spool\failed\`
- ☐ The instrument ID, agent version, and what was expected versus observed

BEFORE SENDING

Delete the `api_token` line from any copy of `config.toml` you share. Treat it as a password.

C4. Upgrading

Configuration, baselines and stored results are preserved across upgrades.

```
pip install --user --upgrade "git+https://github.com/webwebb56/mdqc-py.git"
```

Or run the newer installer over the existing installation. Restart the agent afterwards and confirm the version on the Dashboard. After any upgrade, verify that a run completes end to end before leaving the system unattended.

C5. Data handling and IT review

Summary for an IT or information-security reviewer.

Question	Answer
What leaves the instrument PC?	Derived QC measurements only — peptide retention times, peak areas, quality scores and run metadata. Raw data files are never transmitted. A result file is approximately 25 KB.
Where does it go?	The Mass Dynamics platform endpoint configured at installation, over HTTPS.
How is it authenticated?	A bearer token issued to the customer's own account, stored in <code>config.toml</code> on the instrument PC.
Inbound connections?	None. The agent only makes outbound requests. Its interface is bound to the local loopback address and is not reachable from the network.
What if the network is unavailable?	Results queue locally and upload when connectivity returns.
Can it run with no network at all?	Yes. Omit the cloud configuration. All monitoring works locally; only cross-instrument and longitudinal review on the platform is unavailable.
What is retained locally?	Results under <code>spool\</code> , bounded by count or age (Annex 1), and baselines under <code>gold_standards\</code> .
Third-party components?	Skyline (MacCoss Lab, University of Washington) performs extraction locally. The agent is open source under the MIT licence.

C6. Uninstall

- 1 Stop the agent — Services, or close the foreground window.
- 2 Remove via Add/Remove Programs, or `pip uninstall mdqc`.
- 3 `C:\ProgramData\MassDynamics\QC\` is left in place. Retain it if results or baselines are still needed; delete it for a complete removal.

Annex 1 — Configuration reference

`C:\ProgramData\MassDynamics\QC\config.toml`. Most settings are also available in the Settings page; edit the file directly only where noted.

Key	Default	Effect
[agent]		
log_level	info	Increase to debug when investigating an issue.
[cloud]		
endpoint	Development	Platform environment. Use Production for customer installations.
api_token	—	Bearer token. Omit for local-only. Treat as a password.
[skyline]		
path	auto	Explicit SkylineCmd.exe path if auto-detection fails.
timeout_seconds	900	Extraction timeout. Raise for very large acquisitions.
report_skyr_path	auto	Path to a site-specific report definition.
collapse_transitions_to_peptides	true	Report one row per peptide rather than per transition. Leave enabled.
[watcher]		
stability_window_seconds	60	How long a file must stop changing before it is treated as complete.
[spool]		
completed_retention_count	200	Results retained locally once cloud upload is configured. Set above the busiest expected night.
max_age_days	30	Age limit. With no cloud upload configured this is the only limit applied.
[[instruments]] — one block per instrument		

Key	Default	Effect
<code>id</code>	—	Stable unique name. Keys the baseline; changing it orphans history.
<code>vendor</code>	—	<code>thermo</code> , <code>bruker</code> , <code>sciex</code> , <code>waters</code> or <code>agilent</code> .
<code>watch_path</code>	—	Acquisition folder.
<code>file_pattern</code>	—	For example <code>*.raw</code> or <code>*.d</code> .
<code>[[peptide_classes]]</code> — one block per class		
<code>protein_name</code>	—	Matched case-insensitively against the Skyline Protein column.
<code>purpose</code>	<code>recovery</code>	<code>recovery</code> , <code>digest_efficiency</code> , <code>oxidation</code> , <code>alkylation</code> or <code>custom</code> .
<code>exclude_from_recovery</code>	<code>false</code>	Exclude from the target-recovery figure.
<code>[qc_thresholds]</code> — editable under Settings → QC thresholds ; values and provenance in Annex 4		

Annex 2 — File-naming specification

Markers may appear anywhere in the filename and are matched case-insensitively. Each must be delimited by the start or end of the name, or by `_`, `-`, a space or a full stop. The vendor extension is ignored.

Marker	Accepted forms	Result
Control type	SSC ₀ , SSC	Classified SSC ₀
	QCA , QC_A , QC-A	Classified QC A
	QCB , QC_B , QC-B	Classified QC B
	BLANK , BLK	Classified blank
SPD	200SPD , 200spd , 200-SPD	Selects the matching baseline. Required where more than one method is used.
Dilution	75perc , 75pct , 75%	Recorded for dilution studies. Optional.
Well	A1 ... H12	Recorded. Raises classification confidence.

Worked example

QCB_75perc_2026-07-24_ss_50ng_Manual_200spd_k562_S1-F8_1.d

QC B

75% dilution

200 SPD

well F8

PRECEDENCE AND PITFALLS

Where several control-type markers are present, the order of precedence is SSC₀, QC A, QC B, blank. A marker embedded in a longer word is not matched — 200spdx does not register as an SPD. Custom rules in the configuration take precedence over all built-in markers.

Annex 3 — Payload reference

One JSON file per run. Principal sections:

Section	Contents
<code>run</code>	Filename, acquisition time, instrument, vendor, control type, SPD, dilution, well, plate, classification confidence.
<code>extraction</code>	Backend and version, template, duration, status, any error message.
<code>target_metrics</code>	One entry per peptide: sequence, retention time, peak area, mass error, dot products, peak shape, peptide class.
<code>run_metrics</code>	Target recovery, median retention-time shift, median mass error, digest efficiency.
<code>baseline_context</code>	Which baseline this run was measured against, and a copy of its reference values. Empty where no baseline is active.
<code>comparison_metrics</code>	Ratios and deviations against that baseline, per peptide and per run, plus the thresholds applied.

Full field-level schema is maintained separately for platform integration.

Annex 4 — Default thresholds and provenance

PROVISIONAL

These derive from a single dilution series on one instrument (timsTOF HT, July 2026): QC B at 100%, 75% and 50% across 200, 300 and 500 SPD with eight replicates each, plus sixteen SSC₀ at 200 SPD. The same series is being repeated on further platforms. Values are configurable and expected to change. They are not an instrument acceptance specification.

Setting	Default	Basis
rt_deviation_pct_max	2.0	Retention time varies well under 1% across a batch absent chromatographic change, making it the most reliable indicator of a wrongly integrated peak. Under revision — see the note below.
dot_product_deviation_pct_max	5.0	Expected agreement for a correctly identified target.
dot_product_deviation_pct_suspect	10.0	Together with retention-time deviation, indicates the wrong peak.
peak_area_deviation_pct_suspect	30.0	Large area change accompanying retention-time deviation.
peak_area_deviation_pct_warn	10.0	Approximately a 25% reduction in material reaching the column.
peak_area_deviation_pct_fail	25.0	Approximately a 50% reduction.

Customised ▾

Target extraction

Retention time deviation	2.0	%
--------------------------	-----	---

Dot product deviation —	5.0	%
-------------------------	-----	---

normal

Within this, extraction is considered sound.

Dot product deviation —	10.0	%
-------------------------	------	---

suspect

Beyond this, combined with retention-time drift, flags a wrong peak.

Peak area deviation —	30.0	%
-----------------------	------	---

suspect

Alternative corroborating signal for a wrong peak.

Is the run performing? The response is buffered — a modest change in peak area can reflect a much larger change in material reaching the column.

Warn at	8.0	%
---------	-----	---

recommended 10.0%

Roughly a 25% reduction in material reaching the column.

Fail at	25.0	%
---------	------	---

Roughly a 50% reduction in material reaching the column.

Restore recommended values

Changes apply to the **next run** — no restart needed, unlike cloud settings. The values in force are written into every payload, so any result can be traced back to the thresholds that produced it.

RETENTION-TIME LIMITS ARE BEING RESTATED

CHANGING THEM

Every payload records both the values in force and whether they are still the shipped defaults, so a result can always be traced to the thresholds that produced it, and an instrument that has been tuned is distinguishable from one that has not.

KNOWN BOUNDARY

Against the source dataset, the 75% condition produced a median deviation of -9.5% , marginally inside the 10% warning threshold. A value nearer 8% would capture that condition. Retained at 10% pending data from further platforms, and recorded here so the limitation is visible to anyone tuning these values.

Annex 5 — Commissioning checklist

Site		Instrument ID	
Engineer		Date	
Agent version		Skyline version	

§	Item	Pass	Notes
A1	Prerequisites confirmed; <code>mdqc doctor</code> reports no faults		
A2	Agent installed; starts automatically with the PC		
A3	Instrument ID and watch path configured		
A4	Skyline method and report in place		
A5	Filename convention verified for each QC type in use		
A6	Peptide classes configured; digest efficiency reported		
A7	Test file processed; full target recovery; result written		
A8	Retention limit raised <i>before</i> token entered		
A8	Cloud connected; a result confirmed on the platform		
A9	SSC ₀ series acquired (count: ___) at ___ SPD		
A9	Baseline saved and active; CV% reviewed and acceptable		
B3	Instrument added to the platform module filters		
B	Customer walked through routine operation and re-baselining		

Engineer signature	Customer signature
--------------------	--------------------

Annex 6 — Figure provenance and outstanding captures

Working note for draft preparation; remove before issue.

Captured

Figures 1–13 were captured against MD QC agent v0.5.9–v0.5.13 on a seeded demonstration instance — a real agent serving synthetic but representative data. No production instrument, customer data or credential appears in any of them. The values shown are plausible rather than measured, so any figure a reader might mistake for reference data should be recaptured on a real instrument before issue.

Figure	Section	Notes
1, 7	A2, A5	Genuine command output, reproduced as text rather than as a screenshot so it stays selectable and prints cleanly.
2–6	A3	Setup wizard, steps 1–5.
8, 9	A8	Retention and cloud settings. The token field shows a masked value; no token appears.
10–12	A9	Gold standards. The series deliberately contains one unequilibrated run so the exclusion described in §A9 is visible. Rendered at extended width so all eight peptide columns fit — in normal use the table scrolls horizontally.
13	B2	Dashboard, clipped to the status-at-a-glance region.
Figures 10–12 were recaptured on v0.5.13 following the Evosep review: the earlier set used retention times outside the 200 SPD gradient, and the baseline table has since gained a retention-time standard deviation column.		
18	Annex 4	QC thresholds panel, captured on v0.5.10 with one value off-default so the recommended-value hint and restore action are visible.

Outstanding

The platform figures cannot be produced from a local instance — they need a signed-in Mass Dynamics account with real uploaded data. The first customer installation is the natural opportunity.

Figure	Section	Required content
14	B3	Module details panel showing the instrument selector.
15	B3	The same module populated after the instrument is added.
16	B4	Traffic-light scorecard showing a mix of states.
17	B4	Longitudinal plot over enough history to show a drift.

Capture at 1240 px content width, no customer-identifying information, and quantize to 256 colours before embedding — screenshots of flat UI lose nothing visible and halve in size.

Annex 7 — Revision history

Rev	Date	Author	Change
A	<i>draft</i>	Mass Dynamics	First issue. Covers MD QC agent v0.5.8.

Mass Dynamics · support@massdynamics.com · This document covers the MD QC agent and the Mass Dynamics platform. For the Evosep Lupo, Evosep Eno and Evotip products, refer to the corresponding Evosep documentation.